

Po-Chien Luan

✉ po-chien.luan@epfl.ch | 🎓 Google Scholar | 📍 Lausanne, Switzerland

RESEARCH INTERESTS

• World Models • Human Motion • Social Navigation • State Space Models • 3D Computer Vision

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL)

Ph.D. in Robotics

Sep. 2022 – Present
Lausanne, Switzerland

- Advisor: Prof. Alexandre Alahi
- Collaboration with Honda R&D

National Cheng Kung University (NCKU)

M.S. in Electrical Engineering

Sep. 2018 – Aug. 2020
Taiwan

- Thesis: IFB-PSO for Optimized Realization of an Anthropomorphic Robotic Hand and Electromyographic Hand Gesture Classification Using Lifelong Learning
- Advisor: Prof. Tzuu-Hseng S. Li; Co-advisor: Prof. Ping-Huan Kuo

National Cheng Kung University (NCKU)

B.S. in Electrical Engineering

Sep. 2014 – Jul. 2018
Taiwan

SELECTED PUBLICATIONS

Po-Chien Luan, Zimin Xia, Wuyang Li, Yang Gao, Alexandre Alahi. *Drift-Resistant Navigation World Model with Anchored Epipolar Guidance*. In submission (2026).

Po-Chien Luan, Wuyang Li, Yang Gao, Alexandre Alahi. *Social-Mamba: Socially-Aware Trajectory Forecasting with State-Space Models*. In submission (2026).

Po-Chien Luan*, Yang Gao*, Céline Demonsant, Alexandre Alahi. *Unified Human Localization and Trajectory Prediction with Monocular Vision*. IEEE International Conference on Robotics and Automation (ICRA) (2025).

Ahmad Rahimi*, **Po-Chien Luan***, Yuejiang Liu*, Frano Rajič, Alexandre Alahi. *Sim-to-Real Causal Transfer: A Metric Learning Approach to Causally-Aware Interaction Representations*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2025).

Wuyang Li, Wentao Pan, **Po-Chien Luan**, Yang Gao, Alexandre Alahi. *Stable video infinity: Infinite-length video generation with error recycling*. International Conference on Learning Representations (ICLR) (2026) (Oral).

Yang Gao, **Po-Chien Luan**, Kaouther Messaoud, Lan Feng, Alexandre Alahi. *OmniTraj: Pre-Training on Heterogeneous Data for Adaptive and Zero-Shot Human Trajectory Prediction*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshop on Precognition (2026).

Yang Gao, **Po-Chien Luan**, Alexandre Alahi. *Multi-Transmotion: Pre-trained Model for Human Motion Prediction*. Conference on Robot Learning (CoRL) (2024).

WORK EXPERIENCE

Advanced Institute of Manufacturing with High-tech Innovations (AIM-HI)

Graduate Research Assistant

Jul. 2021 – May 2022
Taiwan

- Advisors: Prof. Her-Terng Yau; Prof. Ping-Huan Kuo
- Published 9 journal papers (3 first-author, 6 co-author)

Taichung Veterans General Hospital Puli Branch

Military Service

Oct. 2020 – Mar. 2021
Taiwan

AWARDS AND HONORS

EPFL–Taiwan Ministry of Education (MoE) Scholarship (2022–2026); one of 5 awards annually
Excellent Teaching Assistant Award, Control Engineering Laboratory (NCKU EE) (2020)
FIRA RoboWorld Cup Androsot: 1st Place (2019)
FIRA RoboWorld Cup HuroCup (Adult Robots All-Round): 1st Place (2018)
KAIST AI World Cup (AI Soccer): 4th Place (2018)

ACADEMIC SERVICE

Teaching Assistantships

Lead Teaching Assistant, Deep Learning for Autonomous Vehicles, EPFL (2026)
Teaching Assistant, Deep Learning for Autonomous Vehicles, EPFL (2023–2024)
Teaching Assistant, Control Engineering Laboratory Course, NCKU (2019)

Reviewer

CVPR; ECCV; IROS; RA-L; T-ITS; TRC; T-IE; T-II; T-CYB

SKILLS

Programming : Python, C/C++, MATLAB, Java

Deep Learning : PyTorch, Hugging Face, Weights & Biases (W&B)

Robotics / Simulation : ROS 1, Gazebo, Webots

Development Tools : Git